DAA Assignment 3 (CO3)

Consolidated Outputs — Questions 1 to 6

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Captured by running DAA_Assignment_3_All_Codes.py. Timing values are specific to this execution environment.

```
DAA ASSIGNMENT 3 — CO3 OUTPUTS
```

Q1A MAX/MIN

```
n= 1 min/max/comparisons= (7, 7, 0)
n= 2 min/max/comparisons= (4, 9, 1)
n= 6 min/max/comparisons= (1, 9, 7)
n= 16 min/max/comparisons= (1, 16, 22)
```

Q1B BINARY SEARCH

```
target 0 =&gt; (0, 9)
target 50 =&gt; (25, 9)
target 98 =&gt; (49, 11)
target 51 =&gt; (-1, 12)
```

Q2 MERGE SORT / QUICK SORT

```
balanced-style merge comparisons 13 quick comparisons 11 partition sizes [(2, 4), (1, 0), (2, 1), (1, 0)] max depth 3 correct T
unbalanced ascending merge comparisons 40 quick comparisons 190 partition sizes [(19, 0), (18, 0), (17, 0), (16, 0), (15, 0), (
```

Q3 KARATSUBA

```
digits 4 correct True standard base products 16 Karatsuba base products 11
digits 8 correct True standard base products 64 Karatsuba base products 33
digits 12 correct True standard base products 91 Karatsuba base products 63
```

Q3 TIMING

```
digits 8 standard 1.1397800e-05 Karatsuba 8.9935333e-06
digits 16 standard 4.7697000e-05 Karatsuba 2.8259800e-05
digits 32 standard 1.5305453e-04 Karatsuba 9.1582267e-05
digits 64 standard 6.1772580e-04 Karatsuba 2.8878907e-04
digits 128 standard 2.7024061e-03 Karatsuba 7.9830567e-04
```

Q4 STRASSEN

```
n 2 correct True conventional multiplications 8 Strassen multiplications 8 times 9.004000e-06 1.945900e-05
n 4 correct True conventional multiplications 64 Strassen multiplications 56 times 7.832000e-06 4.700100e-05
n 8 correct True conventional multiplications 512 Strassen multiplications 392 times 3.774700e-05 2.008810e-04
n 16 correct True conventional multiplications 4096 Strassen multiplications 2744 times 4.524190e-04 1.439232e-03
```

Q5 CLOSEST PAIR

```
n 100 same distance True brute calculations 4950 D&amp;C calculations 134 times 0.00097 0.00030
n 500 same distance True brute calculations 124750 D&amp;C calculations 578 times 0.02543 0.00167
n 1000 same distance True brute calculations 499500 D&amp;C calculations 1116 times 0.10396 0.00374
n 2000 same distance True brute calculations 1999000 D&amp;C calculations 2323 times 0.42225 0.00736
```

Q6A SELECTION

```
kth 5 QuickSelect 14 comparisons 28 Median of Medians 14 comparisons 37
```

Q6B MEET IN THE MIDDLE

```
values [3, 34, 4, 12, 5, 2] target 9 selected indices [2, 4] selected values [4, 5]
```

VALIDATION

```
All validation tests: PASS
```